B.Tech. Computer Science and Engineering

COURSE PLAN: THEORY COURSE

Department :	**Computer Science and Engineering**			
Course Name & code :	**EMBEDDED SYSTEMS CSE 2223**		**Core**	
Semester & branch :	**IV**	**CSE**		
Name of the faculty :	**Dr G Deepa**			
No of contact hours/week:	**L**	**T**	**P**	**C**
	3	**1**	**3**	**4**

Course Outcomes (COs) to PO, PSO, BL Mapping

At the end of this course, the student should be able to:		**No. of Contact Hours**	**Marks**	**Program Outcomes (POs)**	**PSOs**	**BL (Recommended)**
CO1	Explain the architecture and addressing modes of ARM Cortex-M microcontroller	7	14	1, 2, 3, 8, 12	1	3 and Above
CO2	Develop assembly language programs using ARM instruction set.	12	27	2, 3, 4, 5, 8, 9, 12	1	3 and Above
CO3	Compare and contrast different types of stack in ARM	3	7	1, 2, 8, 12	1	3 and Above
CO4	Create the embedded C code for interfacing I/O devices such as LEDs, Seven segment, LCD, keyboard, DC motor, Stepper motor, UART, ADC and DAC.	15	27	1, 2, 3, 4, 5, 8, 9, 10, 12	1	3 and Above
CO5	Demonstrate timer/counter and interrupt programming	11	20	1, 2, 3, 4, 5, 8, 9, 10, 12	1	3 and Above
CO6	Develop an application using ARM Cortex-M microcontroller for interfacing peripherals.	**Self Learning**	**5**	1, 2, 3, 4, 5, 8, 9, 10, 12	**1**	3 and Above
	Total	**48**	**100**			

Course Articulation Matrix

CO	Engineering knowledge	Problem analysis	Design/development of solutions	Investigations of complex problems	Modern tool usage	Engineer and society	Environment and sustainability	Ethics	Individual and team work	Communication	Project management and finance	Life-long learning				
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	1					1				1	3			
CO2		2	1	1	1			1	1			1	3			
CO3	1	2						1				1	3			
CO4	1	2	3	1	1			1	1	1		1	3			
CO5	1	2	3	1	1			1	1	1		1	3			
CO6	1	2	3	1	1			1	1	1		1	3			
Average Articulation Level	**1**	**1.83**	**2.2**	**1**	**1**			**1**	**1**	**1**		**1**	**3**			

ICT Tools used in delivery and assessment

Sl. No	Name of the ICT tool used	Details of how it is used
1	Presentation Slides	To teach in classrooms
2	Online Videos	Link shared via MS Teams, LMS
3	MS Teams	Online communication with students

Mapping of Course Outcomes (COs)/Course Learning Outcomes (CLOs) #

At the end of this course, the student should be able to:		No. of Contact Hours	Marks	Program Outcomes (POs)	Learning Outcomes (LOs)	BL (Recommended)
CLO1	Explain the architecture and addressing modes of ARM Cortex-M microcontroller.	7	14	1, 2, 3, 8, 12	1, 12	3 and above
CLO2	Develop assembly language programs using ARM instruction set.	12	27	2, 3, 4, 5, 8, 9, 12	1	3 and above
CLO3	Compare and contrast different types of stack in ARM	3	7	1, 2, 8, 12	1	3 and above
CLO4	Create the embedded C code for interfacing I/O devices such as LEDs, Seven segment, LCD, keyboard, DC motor, Stepper motor, UART, ADC and DAC.	15	27	1, 2, 3, 4, 5, 8, 9, 10, 12	6, 13	3 and above
CLO5	Demonstrate timer/counter and interrupt programming	11	20	1, 2, 3, 4, 5, 8, 9, 10, 12	6	3 and above
CLO6	Develop an application using ARM CortexM microcontroller for interfacing peripherals.	Self learning	5	1, 2, 3, 4, 5, 8, 9, 10, 12	6, 13	3 and above
	Total	**48**	**100**			

 # Applicable to IET Accredited Courses (modules) Only

Delivery and Assessment Plan of LOs #

Learning Outcome (LO) mapped to the course		**Delivery and assessment Plan**
LO	**LO statement**	
1	***Apply*** knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Some of the knowledge will be at the forefront of the particular subject of study.	Assignments and Written examinations
6	***Apply*** an integrated or systems approach to the solution of complex problems.	Assignments and Written examinations
12	Use practical laboratory and workshop skills to investigate complex problems.	Laboratory Experiments
13	***Select*** and ***apply*** appropriate materials, equipment, engineering technologies and processes, ***recognising their limitations***.	Laboratory Experiments and Assignments

 # Applicable to IET Accredited Programs Only

Assessment Plan

IN – SEMESTER ASSESSMENTS							
S. No.	**Assessment Method**	****Time Duration**	****Marks**	**** Weightage**	**Typology of Questions (Recommended)**	****Schedule**	****Topics Covered**
1	**Assignment 1 (IA1)**	**15 Minutes**	**5**	10 MCQs × ½ = 5	L3 and above	Jan 31 – Feb 11, 2025	L1-L9
2	**Assignment 2 (IA2)**	**15 Minutes**	**5**	10 MCQs × ½ = 5	L3 and above	Feb 17 – Feb 24, 2025	L10-L19
3	**Mid-Term Test**	**90 Minutes**	**30**	**Objective:** 5M 10 MCQs × ½ = 5 marks **Descriptive:** 10 M (2 Questions of 4 marks each +5 Questions of 3 marks each +1 Question of 2 marks)	L3 and above	March 3 – 8, 2025	L1-L27
4	**Assignment 3 (IA3)**	**1 Week**	**5**	Take home assignment 5 Questions * 1 =5	L3 and above	March 17 – March 22, 2025	L24-L34
5	**Assignment (IA4)**	**1 Week**	5	Take home assignment 5 Questions * 1 =5	L3 and above	April 10 – April 21, 2025	L31-L40

END-SEMESTER ASSESSMENT							
1	**Regular/Make–Up Exam**	180 Mins	50	Answer all 5 full questions of 10 marks each. Each question can have 3 parts of 2/3/4/5 marks.	L3 and above		

NOTE: ***Information provided in the table is as per the In-semester assessment plan notified by Associate Director (Academics).***

Lesson Plan

L. No.	Topics	CO Addressed
0	To address OBE philosophy towards NBA and IET accreditation followed by a discussion on Course Plan, Course Outcomes, CO-PO Mapping and Blooms Taxonomy, with prerequisites for the course, if any.	-
1	Introduction to Embedded Systems, Microprocessors and Microcontrollers, an overview of ARM-Cortex-M Architecture: General purpose registers	CO1
2	ARM memory map, Load store instructions in ARM, ARM CPSR, ARM Data format	CO1
3	Pseudo instructions and Directives	CO1
4 (T1)	Introduction to ARM Assembly Programming, Some ARM Addressing modes,	CO1
5	Advanced Indexed Addressing Mode	CO1
6	ADR, LDR and PC Relative addressing	CO1
7	RISC Architecture in ARM	CO1
8 (T2)	Arithmetic Instructions	CO2
9	Logic Instructions	CO2
10	Rotate and Barrel Shifter	CO2
11	Shift and Rotate Instructions	CO2
12 (T3)	Packed and Unpacked BCD	CO2
13	BCD and ASCII conversion	CO2
14	Looping Instructions	CO2
15	Branch Instructions	CO2
16 (T4)	Calling Subroutine and Return	CO2
17	Conditional Execution	CO2
18	Recursion	CO2
19	Conditional Execution	CO2
20 (T5)	Stack and Stack usage in ARM: How stacks accessed, Push and Pop	CO3
21	STM, LDM instructions	CO3
22	Using LDM and STM instructions for the stack	CO3
23	Input/Output (IO) Programming: Pin connect block, Pin function select registers	CO4
24 (T6)	General Purpose Input and Output (GPIO) registers	CO4
25	GPIO configuration, GPIO programming using ARM C language	CO4
26	Interfacing: LEDs	CO4
27	Seven Segment Display	CO4
28 (T7)	LCD	CO4
29	Keyboard	CO4
30	DC motor, Stepper motor	CO4
31	TIMER/COUNTER PROGRAMMING: Timer versus counter	CO5
32 (T8)	Timer Registers	CO5

33	Timer architecture and operation	CO5
34	Timer/counter programming	CO5
35	Nested Vectored Interrupt Controller (NVIC)	CO5
36 (T9)	Nested Vectored Interrupt Controller (NVIC)	CO5
37	external hardware interrupts	CO4
38	IO interrupts	CO4
39	Timer/counter interrupts	CO4
40 (T10)	Serial, ADC and DAC Interfacing: General introduction to serial interfacing	CO4
41	RS232, MAX 232 UART, UART programming	CO4
42	Data acquisition system	CO4
43	Analog to Digital Converter (ADC)	CO4
44 (T11)	ADC registers, Digital to Analog converter (DAC)	CO5
45	DAC registers	CO5
46	ADC and DAC programming	CO5
47	PWM timer and architecture	CO5
48 (T12)	PWM Programming	CO5

Faculty members teaching the course (if multiple sections exist):

Faculty	Section	Faculty	Section
Dr Prashanth Barla	A	Dr G Deepa	C
Mr. Manamohana Krishna	B	Dr Hemalatha S	D

References:

1	Muhammad Ali Mazidi, Sarmad Naimi, Sepehr Naimi, Shujen Chen, ARM Assembly Language Programming & Architecture (2e), MicroDigital Ed, 2016
2	Jonathan W. Valvano., Embedded systems: Introduction to Arm(r) Cortex-M Microcontrollers (5e), Createspace Independent publishing platform, June 2014.
3	Jonathan W. Valvano., Embedded systems: real-time interfacing to ARM Cortex-M microcontrollers (4e), Createspace Independent Publishing Platform, 2017.
4	UM10360, LPC 176x/5x User Manual, NXP Semiconductors, Rev. 4.1, 2016.
5	Toulson and Tim Wilmshurst., Fast and Effective Embedded System Design applying the ARM mbed, Elsevier, 2017.
6	Joseph V., A definitive Guide to ARM Cortex-M3 and Cortex-M4 processors (3e), Elsevier, 2014

Submitted by:

(Signature of the faculty)

Date:

Approved by:

(Signature of HOD)